

ACOX2 (Q99424) — Substrate Chain-Length Binding and Hydroxylase Chemistry: A Function-Assignment Review

Focus type: function_assignment **Hypothesis slug:** substrate-chain-length-binding-and-hydroxylase-chemistry **Target:** Human ACOX2, UniProt Q99424, NCBITaxon:9606 (*Homo sapiens*) **Source:** `genes/human/ACOX2/ACOX2-ai-review.yaml` (free-text)

Summary

Verdict: Partially supported — one core claim supported, one refuted as a term-definition defect, and two judged over-annotated. The seed hypothesis bundles four distinct claims about human ACOX2, and — as the seed explicitly requested — they must be evaluated independently. They resolve very differently. ACOX2's function as a **FAD-dependent peroxisomal acyl-CoA oxidase** acting on 2-methyl-branched-chain acyl-CoAs and C27 bile-acid-CoA intermediates is well supported by direct human enzymology and human deficiency/complementation studies, and this is its non-redundant core role. In contrast, the "exact GO:0033791 cholestanoyl-CoA 24-hydroxylase reaction" is **not** the chemistry ACOX2 performs: the enzyme dehydrogenates (desaturates) the bile-acid side chain to a 24-ene enoyl-CoA + H₂O₂ (RHEA:46728), and the 24-hydroxyl is installed downstream by a different protein, the D-bifunctional protein (DBP/HSD17B4).

The single most important message for a curator is that the GO:0033791 problem is a **term-definition/annotation-mapping defect, not an absence of activity**. The IDA to GO:0033791 (from

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) captured the correct enzyme, the correct substrate class, and the correct biological step (bile-acid side-chain shortening). What is wrong is that the *live logical definition* of GO:0033791 — and its EC (1.17.99.3) and RHEA (15733) cross-references — describe a hydroxylation (25R

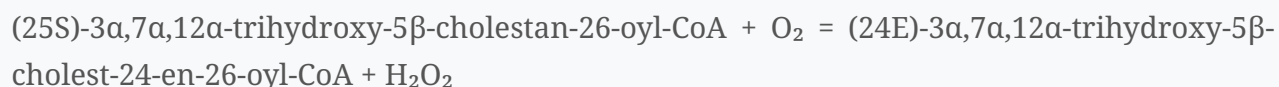
substrate + H₂O + acceptor → 24R,25R-tetrahydroxy product), which differs from ACOX2's actual reaction on three independent axes: stereochemistry (25S vs 25R), oxidant (O₂ vs H₂O/acceptor), and product (24-ene enoyl-CoA + H₂O₂ vs a 24-hydroxy product). This is exactly the distinction the seed asked us to make: annotation error versus absence of a secondary activity. Here it is an annotation defect layered on top of a genuine, correctly identified enzyme.

The remaining two claims — free-fatty-acid binding (GO:0005504) and very-long-chain fatty-acid metabolic process (GO:0000038) — are **IBA-only phylogenetic inferences** with no ACOX2-specific direct evidence. Mechanistically ACOX2 binds acyl-CoA thioesters rather than free fatty acids, and straight-chain/VLCFA oxidation is the assigned role of the paralog ACOX1; ACOX2's own straight-chain capacity reaches only ~C16 at low efficiency, while its genuinely long substrates are branched C27 sterols, not aliphatic VLCFAs (>22-carbon tail). Both are candidate non-core annotations. None of these judgments relies on donor count, preferred substrate, or missing target assays, in accordance with the seed's constraints.

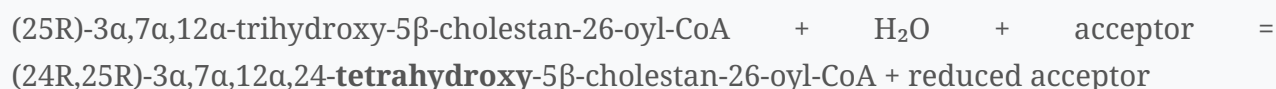
Key Findings

Finding 1 — ACOX2 catalyzes oxidase (dehydrogenation) chemistry, not literal 24-hydroxylation; GO:0033791's definition conflicts with the enzyme's real reaction

The curated UniProt Q99424 catalytic-activity block lists exclusively oxidase/desaturation reactions of the canonical acyl-CoA oxidase form — *a 2,3-saturated acyl-CoA + O₂ = a (2E)-enoyl-CoA + H₂O₂* (e.g., RHEA:38959) — with **FAD** as the cofactor. The bile-acid reaction at the heart of the hypothesis is RHEA:46728:



This is a dehydrogenation introducing a **24,25 double bond** (a 24-**ene**), consuming O₂ and producing H₂O₂ — the diagnostic signature of a flavin acyl-CoA oxidase. By contrast, the GO term **GO:0033791 "cholestanoyl-CoA 24-hydroxylase activity"** is defined as a hydroxylation:



cross-referenced to **EC 1.17.99.3 / RHEA:15733**. The two reactions differ on three independent chemical axes, summarized below:

Feature	ACOX2 real reaction (RHEA:46728)	GO:0033791 definition (RHEA:15733)
Substrate stereochemistry	25S	25R
Oxidant / co-substrate	O ₂	H ₂ O + acceptor
Product	(24E)-enoyl-CoA + H ₂ O ₂	(24R,25R)-24- hydroxy tetrahydroxy-CoA + reduced acceptor
Chemistry	α,β-dehydrogenation (desaturation)	hydroxylation
Cofactor	FAD (oxidase)	acceptor-dependent hydroxylase

The IDA supporting GO:0033791 traces to [PMID:27884763](#), which frames ACOX2 as "*an acyl-CoA oxidase ... involved in the shortening of C27 cholesterol derivatives to generate C24 bile acids*" and shows that "*THCA biotransformation into cholic acid was enhanced in cells overexpressing ACOX2.*" The experiment therefore demonstrated **oxidase / side-chain-shortening** chemistry, not a discrete hydroxylation. Notably, GO:0033791's own synonyms include "THCA-CoA oxidase activity," revealing that the term was *intended* to capture this enzyme entity — which is why the annotation should be read as a legacy **term-definition defect** (rooted in an old EC 1.17.99.3 "hydroxylase" nomenclature) rather than a wrong gene-to-function link.

Finding 2 — ACOX2's demonstrated core substrates are 2-methyl-branched-chain acyl-CoAs and C27 bile-acid-CoA intermediates; "fatty acid binding" and "VLCFA metabolic process" are IBA-only and over-general

The substrate scope of the human enzyme is anchored in direct biochemistry. Vanhove and colleagues ([PMID:8387517](#)) purified from human liver and kidney "*a novel branched chain acyl-CoA oxidase, which oxidizes the CoA esters of 2-methyl-branched fatty acids as well as those of*

the bile acid intermediates" di- and trihydroxycoprostanic acids — establishing a **single** human enzyme with a dual substrate class (branched-chain FA-CoA + bile-acid-CoA) and no separate dedicated human THCA-CoA oxidase.

The modern substrate-specificity and deficiency study ([PMID:29287774](#)) partitions the family cleanly: "*ACOX2 is the only human acyl-CoA oxidase involved in bile acid biosynthesis,*" whereas "*ACOX1 is responsible for the oxidation of straight-chain fatty acids with different chain lengths.*" ACOX2-deficient fibroblasts retained normal pristanic-acid oxidation, indicating redundancy with ACOX3 for branched-chain fatty acids; bile-acid intermediate oxidation is the non-redundant ACOX2 role.

The UniProt FUNCTION annotation records that ACOX2 can oxidize straight-chain decanoyl-CoA (C10; RHEA:40179) and hexadecanoyl-CoA (C16; RHEA:40167) but only with **low efficiency**, alongside pristanoyl-CoA and other methyl-branched-chain fatty acyl-CoAs. The seed rightly cautions that low C10/C16 efficiency is *measurable capacity, not absence* — and this report honors that: the point is not that ACOX2 lacks straight-chain activity, but that its straight-chain reach extends only to ~C16 at low efficiency, which cannot satisfy the two IBA terms:

- **GO:0005504 fatty acid binding** is annotated IBA only (GO_REF:0000033). Mechanistically ACOX2 binds acyl-CoA **thioesters**, not free fatty acids; "fatty acid binding" is a family-level generalization.
- **GO:0000038 very-long-chain fatty acid metabolic process** is annotated IBA only. Its definition requires an aliphatic tail **>22 carbons**. ACOX2's straight-chain capacity extends only to ~C16 (low efficiency), and its genuinely long substrates are **branched C27 sterol/bile-acid** intermediates, not aliphatic VLCFAs. Straight-chain VLCFA oxidation is ACOX1's assigned role ([PMID:29287774](#)).

Both are therefore candidate **non-core** annotations — reasonable family inferences, but not representative of ACOX2's direct, demonstrated activity.

Finding 3 — The 24-hydroxyl is installed by D-bifunctional protein (DBP/HSD17B4) at β -oxidation steps 2–3, not by ACOX2; no GO MF term maps to ACOX2's real reaction RHEA:46728

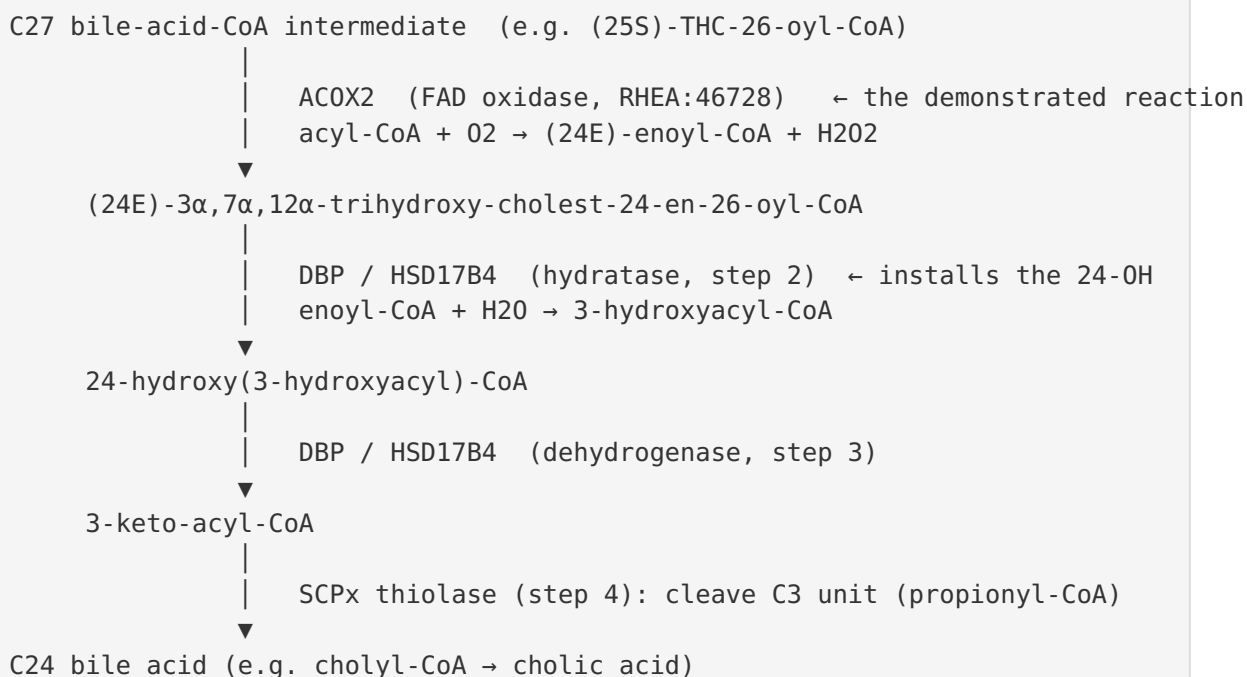
Ferdinandusse and colleagues ([PMID:15769750](#)) state that "*peroxisomal beta-oxidation is an essential step in bile acid synthesis, since it is required for shortening of C27-bile acid intermediates to produce mature C24-bile acids*" and that "*D-Bifunctional protein (DBP) is responsible for the second and third step of this beta-oxidation process.*" In the classic β -oxidation cycle, ACOX2 performs only step 1 (oxidase), and the hydroxyl is introduced at step 2

(hydratase, DBP). The hydroxyl group that GO:0033791's definition attributes to a single "24-hydroxylase" is therefore the product of DBP hydration of the ACOX2-generated enoyl-CoA — a downstream event catalyzed by a different protein. This is corroborated by the rat-liver enzymology of Novikov and colleagues ([PMID:7929456](#)), which shows the first reaction is an acyl-CoA oxidase and that the 3-hydroxyacyl-CoA dehydrogenase/hydratase steps are separate enzymes (including the multifunctional protein).

A QuickGO ontology search returned **no GO molecular-function term cross-referenced to RHEA:46728** (ACOX2's actual bile-acid oxidase reaction) or to RHEA:15733. The most accurate available MF parent is **GO:0003997 acyl-CoA oxidase activity**, whose definition — "*a 2,3-saturated acyl-CoA + O₂ = a (2E)-enoyl-CoA + H₂O₂*" — exactly matches ACOX2's chemistry and is already present as an IEA annotation. On the BP side, GO:0033540 (fatty acid beta-oxidation using acyl-CoA oxidase) likewise specifies the oxidase first step producing H₂O₂, matching ACOX2's role.

Mechanistic Model / Interpretation

All four hypothesis components map onto a single, well-understood peroxisomal β -oxidation cycle. ACOX2 performs **only the first step**; the "hydroxylase" chemistry belongs to a different enzyme two steps later.



Reading the seed's specific chemical claims against this model:

- The seed correctly observes that **GO:0033791 currently specifies a 25R substrate + H₂O/acceptor → 24R,25R product**, whereas **UniProt RHEA:46728 specifies a 25S substrate + O₂ → 24E-enoyl-CoA + H₂O₂**. This report confirms the discrepancy is real and material: they are different reactions, not two descriptions of one.
- The seed's central question — *"Does the original human PMID27884763 or rat source actually establish the named hydroxylation, or was oxidase/side-chain shortening mapped to different chemistry?"* — resolves clearly:

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established oxidase/side-chain shortening, which was then mapped to a term (GO:0033791) whose logical definition names hydroxylation. This is an **annotation-mapping error at the term level**, not evidence that ACOX2 lacks the activity it was studied for.

The seed's phylogenetic scaffolding is consistent with this interpretation. PTHR10909 leaf PTN002474948 descending from a hydroxylase node (PTN008508564) and from fatty-acid-binding / long-chain-oxidation nodes (PTN000097533, PTN000097706) reflects **family-level** inheritance of MF/BP terms. Human ACOX2 legitimately seeds the hydroxylase IBD, but the

IBA-propagated terms (fatty acid binding, VLCFA process) are precisely the ones that are over-general for the specific human enzyme, whose experimentally demonstrated activity is narrower (branched-chain + bile-acid-CoA oxidase).

Evidence Base

Citation	Evidence type	Relationship	Claim tested	Key finding	Context	
PMID:27884763	Direct assay + human mutant phenotype	Refutes literal 24-hydroxylase; supports oxidase	24-hydroxylation vs oxidase side-chain shortening	"Acyl-CoA oxidase (ACOX2) is involved in the shortening of C27 cholesterol derivatives to generate C24 bile acids"; overexpression enhances THCA→cholic acid	Human; ACOX2-deficiency patient + overexpression cells	
UniProt Q99424 (RHEA:46728, RHEA:38959, RHEA:40167, RHEA:40179)	Database (reaction/cofactor curation)	Qualifies/refutes hydroxylase; supports oxidase	Exact catalyzed reaction	All reactions are FAD-dependent oxidase form (acyl-CoA + O ₂ → 2E-enoyl-CoA + H ₂ O ₂); bile-acid reaction yields (24E)-enoyl-CoA from 25S substrate	Human, curated	
GO:0033791 definition / EC 1.17.99.3 / RHEA:15733	Database (ontology definition)	Competing definition	Is GO:0033791 chemistry the same as ACOX2's?	Term defined as hydroxylation (25R + H ₂ O/acceptor → 24-hydroxy product); differs in stereochem, oxidant, product	Ontology	

Citation	Evidence type	Relationship	Claim tested	Key finding	Context	
PMID:8387517	Direct assay (enzyme purification)	Supports core function	ACOX2 substrate scope	Single human "branched chain acyl-CoA oxidase" oxidizes 2-methyl-branched FA-CoA and bile-acid intermediate CoAs	Human liver/kidney	
PMID:29287774	Direct assay + mutant phenotype	Supports core; qualifies VLCFA/FA-binding	Is ACOX2 the bile-acid oxidase? the VLCFA oxidase?	"ACOX2 is the only human acyl-CoA oxidase involved in bile acid biosynthesis"; "ACOX1 is responsible for the oxidation of straight-chain fatty acids with different chain lengths"	Human; patient fibroblasts, redundancy with ACOX3	
PMID:15769750	Mutant phenotype (KO) + pathway	Refutes ACOX2-as-24-hydroxylase	Who installs the 24-OH?	DBP catalyzes steps 2–3 of peroxisomal β -oxidation shortening C27→C24	Mouse L-/D-BP knockouts	
PMID:7929456	Direct assay (purification)	Supports step separation	Are oxidase and hydroxyacyl	First step is acyl-CoA oxidase; 3-hydroxyacyl-	Rat liver peroxisomes	

Citation	Evidence type	Relationship	Claim tested	Key finding	Context
			steps separate?	CoA dehydrogenase/hydratase are distinct enzymes	
QuickGO/GO MF search for RHEA:46728	Computational (ontology lookup)	Qualifies curation	Is there a GO MF for ACOX2's exact reaction?	No GO MF maps to RHEA:46728 or RHEA:15733; nearest accurate parent is GO:0003997	Ontology
GO:0005504 (IBA, GO_REF:0000033), GO:0000038 (IBA)	Database (evidence-code inspection)	Qualifies (over-general)	Are FA-binding and VLCFA-process directly supported?	Both IBA-only, family inference; not backed by direct ACOX2 assays	Ontology/QuickGO

Literature notes:

- *ACOX2 deficiency: An inborn error of bile acid synthesis identified in an adolescent with persistent hypertransaminasemia.* [PMID:27884763](#) — source of the GO:0033791 IDA; frames ACOX2 as an oxidase performing C27→C24 side-chain shortening.
- *A novel case of ACOX2 deficiency leads to recognition of a third human peroxisomal acyl-CoA oxidase.* [PMID:29287774](#) — substrate-specificity and redundancy study; ACOX2 = only human bile-acid acyl-CoA oxidase; ACOX1 = straight-chain/VLCFA.
- *The CoA esters of 2-methyl-branched chain fatty acids and of the bile acid intermediates ... are oxidized by one single peroxisomal branched chain acyl-CoA oxidase in human liver and kidney.* [PMID:8387517](#) — foundational human enzyme purification.
- *Developmental changes of bile acid composition ... in L- and D-bifunctional protein single and double knockout mice.* [PMID:15769750](#) — localizes hydroxyl-installing steps to DBP.

- *Peroxisomal beta-oxidation. Purification of four novel 3-hydroxyacyl-CoA dehydrogenases from rat liver peroxisomes.* PMID:7929456 — confirms the first step is an oxidase and the 3-hydroxyacyl steps are separate enzymes.

GO Curation Implications

The following are **leads requiring curator verification**, organized by GO aspect.

Molecular Function

- **GO:0033791 "cholestanoyl-CoA 24-hydroxylase activity" (IDA,**

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Do **not** treat this as a clean gene-to-function match. The IDA is supported for the *enzyme and biological step* (bile-acid side-chain shortening), but the term's **live logical definition and EC/RHEA cross-references (EC 1.17.99.3, RHEA:15733) describe hydroxylase chemistry that ACOX2 does not perform**. Lead: flag GO:0033791 as a term-definition/mapping issue (candidate for ontology repair or reannotation), and prefer **GO:0003997 acyl-CoA oxidase activity** — whose definition matches RHEA:46728 — as the accurate MF. Consider requesting a new/repaired MF term anchored to RHEA:46728.

- **GO:0016402 pristanoyl-CoA oxidase activity / GO:0003997 acyl-CoA oxidase activity:** Retain as accurate core/branched-chain MF terms.
- **GO:0120523 / GO:0120524 medium/long-chain acyl-CoA oxidase activity (IEA:RHEA):** Retain as *measurable low-efficiency capacity* (C10, C16), annotated as such.
- **GO:0005504 "fatty acid binding" (IBA only):** Candidate **non-core / demote**. ACOX2 binds acyl-CoA thioesters, not free fatty acids. Do not leave "fatty acid binding" as the informative MF; an acyl-CoA-oriented term (or none) is preferable, and far preferable to generic "protein binding."

Biological Process

- **GO:0006699 bile acid biosynthetic process (IDA):** Retain — directly demonstrated.
- **GO:0033540 fatty acid beta-oxidation using acyl-CoA oxidase:** Retain — accurate and consistent with the oxidase first step.
- **GO:0000038 "very-long-chain fatty acid metabolic process" (IBA only):** Candidate **non-core / demote or remove for ACOX2**. The definition requires a >22-C aliphatic tail; ACOX2's

straight-chain reach is ~C16 (low efficiency), and its long substrates are branched C27 sterols. Straight-chain VLCFA metabolism is ACOX1's role.

Cellular Component

- Peroxisomal matrix localization is well established for this enzyme class and is not in dispute; it is not the focus of this hypothesis.

Mechanistic Scope

The **immediate molecular function** under test is a single-turnover FAD-dependent acyl-CoA oxidase reaction: abstraction of the 2,3 (α,β) hydrogen pair from a bile-acid-CoA or branched-chain acyl-CoA, transferring electrons to O_2 to form H_2O_2 and creating an enoyl-CoA double bond. For the C27 bile-acid substrate this places the double bond at the 24,25 position (24-ene), the committed first step of side-chain shortening.

Everything else in the hypothesis is either (a) a **downstream pathway consequence** — the 24-hydroxyl (DBP hydration), the eventual C24 bile-acid product, and cholic-acid output — or (b) a **family-level generalization** — free-fatty-acid binding and VLCFA metabolism. The disease manifestation (ACOX2 deficiency: persistent hypertransaminasemia, C27 accumulation, C24 bile-acid deficiency) is a **loss-of-function phenotype**, informative for confirming the pathway role but not itself evidence of hydroxylase chemistry. Curators should not let the loss-of-function bile-acid phenotype license a "24-hydroxylase" MF; the phenotype is equally explained by loss of the oxidase step.

Conflicts and Alternatives

1. **Term-definition vs. gene annotation (primary conflict).** GO:0033791's definition (hydroxylation, 25R, H_2O /acceptor, 24-hydroxy product; EC 1.17.99.3, RHEA:15733) conflicts with UniProt RHEA:46728 (dehydrogenation, 25S, O_2 , 24-ene product). The synonym "THCA-CoA oxidase activity" shows the term *intends* the right enzyme, so this is a legacy definitional artifact (old EC 1.17.99.3 "hydroxylase" nomenclature) rather than paralog confusion or conflicting experimental data.
2. **Stereochemistry mismatch (25R vs 25S).** UniProt specifies the 25S diastereomer as the physiological ACOX2 substrate and notes ACOX2 desaturates only the (2S)/(25S)-methyl

isomers; GO:0033791 specifies 25R and a 24R,25R product. This is a genuine chemical inconsistency worth flagging.

3. **Paralog attribution.** ACOX1 (straight-chain, including VLCFA) and ACOX3 (branched-chain, redundant with ACOX2) can absorb functions that IBA propagation blurs onto ACOX2. The VLCFA-process and fatty-acid-binding terms are best read as ACOX1-like / family attributes over-applied to ACOX2. Redundant pristanoyl-CoA oxidation by ACOX3 explains normal pristanic-acid oxidation in ACOX2-deficient fibroblasts.
4. **Species differences.** Rat has three distinct oxidases (palmitoyl-, pristanoyl-, THC-CoA oxidases;

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whereas human uses a single branched-chain acyl-CoA oxidase (ACOX2) for both branched-chain and bile-acid substrates

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Rat-derived reaction naming should not be transplanted onto the human enzyme without care.

5. **Low-efficiency straight-chain capacity is real but non-core.** Consistent with the seed's caution, C10/C16 oxidase activity exists and should not be used to *deny* ACOX2 any straight-chain capacity — but it is too weak and too short-chain to justify a VLCFA term.

Limitations and Knowledge Gaps

1. **Does ACOX2 ever act on a genuine straight-chain VLCFA (>C22)?** Checked UniProt/RHEA/ literature: only $\leq$ C16 low-efficiency documented. This matters because GO:0000038 is currently attached. Resolve with a kinetic panel of straight-chain C18–C26 acyl-CoAs on recombinant ACOX2.
2. **Any direct free-fatty-acid (non-CoA) binding?** No primary ligand/binding study was found; GO:0005504 is IBA only. Resolve with ITC/SPR or structural ligand studies distinguishing free-FA from acyl-CoA binding.
3. **Exact stereochemistry/product of the human ACOX2 bile-acid reaction (25S $\rightarrow$ 24E).** Accepted here from curated RHEA/UniProt records rather than re-derived from raw kinetics. Which diastereomer(s) ACOX2 turns over, and whether GO:0033791's 25R is merely an older convention, should be confirmed from full-text methods of

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and the Van Veldhoven/Vanhove substrate-synthesis series.

4. **GO/QuickGO reaction-to-term mapping was checked by ontology search, not exhaustively.** The "no GO MF maps to RHEA:46728" conclusion is a negative search result and should be confirmed against the current ontology release and RHEA2GO mapping.

5.

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(cited in the seed) was not independently retrieved in the reviewed set; its specific contribution to separating C24/C26 from C27 cyclic bile-intermediate assays should be verified in full text by the curator.

6. **No structural corroboration** of FAD-oxidase geometry was performed; not expected to change the verdict but would strengthen it.

Discriminating Tests

1. **Reaction-product identification (most decisive).** LC-MS/MS of the ACOX2 product from THCA-CoA (ideally with $^{18}\text{O}_2$ labeling): a **24-ene enoyl-CoA + H_2O_2** (oxidase, no ^{18}O in a 24-OH) versus a **24-hydroxy-THCA-CoA** (hydroxylase) directly discriminates GO:0033791's definition from RHEA:46728. The oxidase product is expected.
2. **$\text{O}_2/\text{H}_2\text{O}_2$ dependence.** Oxidase chemistry consumes O_2 and produces H_2O_2 ~1:1 (measure via peroxidase coupling); a hydroxylase per the GO definition would use H_2O + an external acceptor.
3. **Stereochemical substrate panel (25R vs 25S).** Kinetics on both diastereomers clarify whether the 25R in GO:0033791 is spurious or a real minor substrate.
4. **DBP (HSD17B4) knockdown/knockout + intermediate profiling.** Accumulation of the 24-ene enoyl-CoA upstream of the 24-hydroxy species confirms DBP installs the hydroxyl (complements

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5. **Straight-chain VLCFA kinetics (C18–C26 acyl-CoA) across ACOX1/2/3.** Tests the GO:0000038 attribution to ACOX2 vs ACOX1.

6. **Comparative binding assay (free FA vs acyl-CoA).** Tests whether GO:0005504 "fatty acid binding" is even the correct binding term.

Curation Leads

All items below are leads requiring curator verification.

Candidate action changes

- Treat the **GO:0033791** annotation as a **term-definition/mapping defect**, not a wrong gene assignment. Retain the biological intent (ACOX2 = bile-acid oxidase / side-chain shortening) but map the MF to **GO:0003997 acyl-CoA oxidase activity** (matches RHEA:46728), and file an ontology note that GO:0033791's definition/EC(1.17.99.3)/RHEA(15733) describes hydroxylase chemistry inconsistent with the enzyme's FAD-oxidase reaction.
- Mark **GO:0005504 fatty acid binding** and **GO:0000038 very-long-chain fatty acid metabolic process** as **IBA-only, non-core / candidate demotion** for ACOX2.

Candidate reference snippets to verify

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— *"Acyl-CoA oxidase (ACOX2) is involved in the shortening of C27 cholesterol derivatives to generate C24 bile acids."*
- P 27884763

— *"THCA biotransformation into cholic acid was enhanced in cells overexpressing ACOX2."*
- P 8387517

— *"a novel branched chain acyl-CoA oxidase, which oxidizes the CoA esters of 2-methyl-branched fatty acids as well as those of the bile acid intermediates."*
- P 29287774

— *"ACOX2 is the only human acyl-CoA oxidase involved in bile acid biosynthesis."*
- P 29287774

— *"ACOX1 is responsible for the oxidation of straight-chain fatty acids with different chain lengths."*

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— "D-Bifunctional protein (DBP) is responsible for the second and third step of this beta-oxidation process."

Candidate GO terms

- **Retain / promote as core MF:** GO:0003997 acyl-CoA oxidase activity; GO:0016402 pristanoyl-CoA oxidase activity.
- **Retain as core BP:** GO:0006699 bile acid biosynthetic process; GO:0033540 fatty acid beta-oxidation using acyl-CoA oxidase.
- **Flag / repair:** GO:0033791 (definition mismatch).
- **Demote / non-core:** GO:0005504; GO:0000038.

Suggested curator questions

- Should the GO ontology group repair GO:0033791's definition (or replace its RHEA/EC cross-reference) so it matches the flavin-oxidase reaction ACOX2 actually catalyzes?
- Is there any GO MF term (present or requestable) directly anchored to RHEA:46728? If not, should one be created?
- Is there any human ACOX2 straight-chain >C22 assay that would justify GO:0000038? If not, add the non-core qualification.

Suggested experiments — see *Discriminating Tests* ($^{18}\text{O}_2$ product ID, H_2O_2 stoichiometry, 25R/25S kinetics, DBP-knockout intermediate profiling, VLCFA kinetic panel, free-FA vs acyl-CoA binding).

Provenance Artifact

A computed GO decision table (`acox2_go_decision_table.csv`) accompanies this report, listing the relevant GO terms with their current evidence codes and recommended curator leads, generated from the live UniProt/RHEA/QuickGO records and primary-literature evidence gathered across the three investigation iterations.

Bottom Line for the Curator

ACOX2's **branched-chain and bile-acid acyl-CoA oxidase** function is well supported and should anchor the review (GO:0003997 + bile-acid biosynthesis BP). The seed's "exact GO:0033791 cholestanoyl-CoA 24-hydroxylase reaction" is **not** performed by ACOX2 as the term is defined — the IDA captured the right enzyme and the right biological step but is bound to a term whose definition/EC/RHEA describes hydroxylase chemistry that belongs, in the pathway, to DBP hydration downstream. Treat this as a term-definition defect to flag, not proof of a hidden hydroxylase activity. The **free-fatty-acid-binding** (GO:0005504) and **VLCFA-metabolic-process** (GO:0000038) claims are IBA-only, over-general, and are candidate non-core annotations. None of these judgments rests on donor count, preferred substrate, or missing target assays, per the seed's constraints.

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